INTERNSHIP REPORT

Machine Learning Research and Plant Disease Classification

Internship at Authentic Four Technology

Submitted By

Md Monir Hossain

Student ID: 0222220005101165

Section: C

Session: Spring 2026

Department of Computer Science and Engineering
Premier University, Chattogram, Bangladesh

Organization:	Authentic Four Technology
Faculty Supervisor:	Mohammad Fahim Foisal
Internship Area:	Machine Learning Research
Research Topic:	Potato Leaf Disease Classification

Internship Period: [Starting Date] – [Ending Date]

Submission Date: [Submission Date]

Declaration

I, Md Monir Hossain, declare that this internship report, titled *"Machine Learning Research and Plant Disease Classification"*, has been prepared based on the machine learning research activities I carried out during my internship at Authentic Four Technology.

The report is submitted as an individual academic work. The technical results presented in this report are based on the dataset, model experiments, and evaluation outputs described herein. External sources used in this report have been acknowledged in the References section.

Md Monir Hossain
Student ID: 0222220005101165
Department of Computer Science and Engineering
Premier University, Chattogram

Acknowledgement

I would like to express my gratitude to Authentic Four Technology for providing me with the opportunity to undertake an internship focused on machine learning research.

I am especially thankful to my industry supervisor, Mohammad Fahim Foisal, for his supervision and guidance during the internship.

I also acknowledge the academic support of the Department of Computer Science and Engineering, Premier University, Chattogram.

Finally, I appreciate the developers and researchers who make open-source machine learning frameworks and datasets available for educational and research purposes.

Executive Summary

This report summarizes my individual internship experience at Authentic Four Technology, focusing on machine learning research and development.

The technical work involved classifying potato leaf diseases using a filtered subset of the PlantVillage dataset, containing three classes: Early Blight, Late Blight, and Healthy.

Two deep learning models were implemented and compared: a Custom Convolutional Neural Network (CNN) and MobileNetV2 using transfer learning and fine-tuning.

The dataset contained 2,152 images, with both models evaluated on the same 323-image test set. The Custom CNN achieved 94.12% accuracy, while MobileNetV2 achieved 97.52%, demonstrating better overall classification performance.

The report covers the organization, internship activities, technologies, technical findings, challenges, proposed improvements, and professional learning outcomes.

Contents

Declaration i

Acknowledgement i

Executive Summary i

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 Internship Overview . . . 1
- 1.3 Objectives of the Internship Work . . . 1
- 1.4 Scope of the Report . . . 2

2 Company Profile 2
- 2.1 Organization Overview . . . 2
- 2.2 Internship Environment . . . 2
- 2.3 Internship Supervision . . . 2
- 2.4 Official Website . . . 3

3 Organizational Structure 3
- 3.1 Reporting and Supervision . . . 3
- 3.2 My Role and Responsibilities . . . 3

4 Technologies Used 4
- 4.1 Programming Language and Environment . . . 4
- 4.2 Libraries and Frameworks . . . 4
- 4.3 Deep Learning Models . . . 4
  - 4.3.1 Custom Convolutional Neural Network . . . 4
  - 4.3.2 MobileNetV2 . . . 4
- 4.4 Dataset and Input Configuration . . . 4

5 Internship Activities 5
- 5.1 Research Problem Selection . . . 5
- 5.2 Dataset Collection and Preparation . . . 5
  - 5.2.1 Sample Dataset Image . . . 5
- 5.3 Image Preprocessing . . . 6
- 5.4 Model Development and Training . . . 7
  - 5.4.1 Custom CNN Implementation . . . 7
  - 5.4.2 MobileNetV2 Implementation . . . 7
- 5.5 Training and Validation Performance . . . 7
  - 5.5.1 Custom CNN Training Performance . . . 8
  - 5.5.2 MobileNetV2 Training Performance . . . 8
- 5.6 Model Evaluation . . . 9
  - 5.6.1 Overall Performance Comparison . . . 10
  - 5.6.2 Confusion Matrix Analysis . . . 10
  - 5.6.3 ROC Curve Analysis . . . 12
  - 5.6.4 Class-wise Performance . . . 13
  - 5.6.5 AUC Comparison . . . 13

6 Technical Learning **13**
6.1 Image Preprocessing 13
6.2 CNN Architecture and Feature Extraction 14
6.3 Transfer Learning and Fine-Tuning 14
6.4 Model Evaluation 14
6.5 Experiment Tracking and Reproducibility 14

7 Problems Observed **14**
7.1 Class Imbalance 14
7.2 Confusion Between Similar Classes 14
7.3 Limited Real-World Diversity 15
7.4 Risk of Overfitting 15
7.5 Metric Consistency and Result Verification 15
7.6 Limited Scope 15

8 Proposed Solutions **15**
8.1 Addressing Class Imbalance 15
8.2 Improving Generalization 15
8.3 Improving Evaluation Reliability 16
8.4 Further Model Investigation 16
8.5 Explainability and Deployment 16

9 Reflection on Professional Skills **16**
9.1 Research and Analytical Skills 16
9.2 Problem-Solving Skills 16
9.3 Technical Communication 16
9.4 Time Management and Independent Work 17
9.5 Professional Conduct and Future Development 17

10 Conclusion **17**

11 References **18**

List of Tables

1 Internship Information 1
2 Technologies and Their Roles 4
3 Distribution of the Filtered Potato Leaf Dataset 5
4 Dataset Split 6
5 Custom CNN Architecture 7
6 Corrected Overall Model Comparison 10
7 Custom CNN Classification Metrics 13
8 MobileNetV2 Classification Metrics 13
9 Recorded One-vs-Rest AUC Scores 13

List of Figures

1 Internship reporting relationship. 3
2 Sample potato leaf image from the dataset. 6
3 Custom CNN training and validation accuracy. 8
4 Custom CNN training and validation loss. 8
5 MobileNetV2 training and validation accuracy. 9
6 MobileNetV2 training and validation loss. 9
7 Confusion matrix of the Custom CNN. 11
8 Confusion matrix of MobileNetV2. 11
9 One-vs-rest ROC curves for the Custom CNN. 12
10 One-vs-rest ROC curves for MobileNetV2. 12

1 Introduction

1.1 Background

Agriculture is an important area where timely identification of plant diseases can support crop management and reduce potential losses.

Disease identification is commonly performed through visual inspection. Although visual inspection is useful, it can require experience and may be time-consuming when many plants need to be examined.

Computer vision and deep learning offer methods for analysing plant leaf images and recognizing visual patterns associated with disease.

Convolutional Neural Networks (CNNs), in particular, can learn image features such as spots, discoloration, texture, and shape patterns. Transfer learning can also make use of features learned from large image datasets.

During my internship at Authentic Four Technology, I worked on a machine learning research task involving image-based potato leaf disease classification.

1.2 Internship Overview

This was an individual internship research activity. My assigned technical focus was to investigate and compare deep learning approaches for classifying potato leaf images.

The work involved preparing a dataset, implementing two models, training and fine-tuning them, evaluating their predictions, and documenting the findings.

Table 1: Internship Information

Item	**Details**
Intern	Md Monir Hossain
Student ID	0222220005101165
Organization	Authentic Four Technology
Industry Supervisor	Mohammad Fahim Foisal
Internship Area	Machine Learning Research
Research Task	Potato Leaf Disease Classification
Nature of Work	Individual research and implementation
Internship Period	[Starting Date] – [Ending Date]

1.3 Objectives of the Internship Work

The main objective was to gain practical experience in applying deep learning to an image-classification problem.

The specific objectives were:

1. Understand and prepare a labelled plant-leaf image dataset.

2. Apply image resizing, normalization, and data splitting.
3. Build a Custom CNN architecture and train it from scratch.
4. Apply transfer learning using MobileNetV2 pretrained on ImageNet.
5. Fine-tune selected MobileNetV2 layers.
6. Evaluate models using classification metrics and visualizations.
7. Compare the models and identify limitations.
8. Document the research findings.

1.4 Scope of the Report

This report focuses on the internship experience and the machine learning research activity.

It does not claim that the experimental model was deployed as a production agricultural diagnosis system.

The evaluation was conducted on a held-out subset of PlantVillage images, and independent field validation was not part of the reported experiment.

2 Company Profile

2.1 Organization Overview

Authentic Four Technology is the organization where I completed my internship as part of my academic and professional development.

The organization maintains an official website at https://authenticfour.com/, which serves as its online presence for sharing information about the company and its activities.

My internship focused on machine learning research, particularly image classification and deep learning applications. During this period, I worked on a research task involving potato leaf disease classification.

2.2 Internship Environment

The internship provided an opportunity to apply academic knowledge to a practical machine learning research problem.

My assigned work involved dataset preparation, model development, transfer learning, fine-tuning, performance evaluation, and technical documentation.

The primary technical activity was the implementation and comparison of a Custom Convolutional Neural Network (CNN) and MobileNetV2 for classifying potato leaf images.

2.3 Internship Supervision

The internship was conducted under the supervision of Imtiaz Uddin Riad, who served as my industry supervisor at Authentic Four Technology.

Mohammad Fahim Foisal served as my faculty internship supervisor, providing academic guidance and support throughout the internship.

Both supervisory roles contributed to the academic and professional context of the internship.

2.4 Official Website

The official website of Authentic Four Technology is:

https://authenticfour.com/

3 Organizational Structure

3.1 Reporting and Supervision

During the internship, my industry supervisor was Mohammad Fahim Foisal.

The available internship information confirms this supervisory relationship. The names and positions of other company officials or departments were not provided.

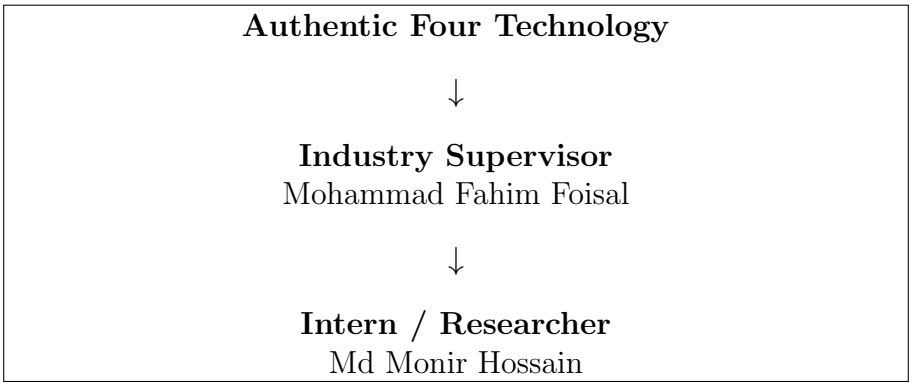


Figure 1: Internship reporting relationship.

3.2 My Role and Responsibilities

As an individual machine learning research intern, my responsibilities included:

- Studying the potato leaf disease classification problem.
- Preparing and organizing the image dataset.
- Implementing and training the selected models.
- Monitoring training and validation performance.
- Evaluating model predictions.
- Comparing experimental results.
- Documenting research findings and limitations.
- Reporting research progress to the industry supervisor.

4 Technologies Used

4.1 Programming Language and Environment

Python was used as the primary programming language for the machine learning workflow.

The experiments used a notebook-based development workflow. Where applicable, the project was executed in Google Colab.

4.2 Libraries and Frameworks

Table 2: Technologies and Their Roles

Technology	Purpose
Python	Main programming language.
TensorFlow / Keras	Building and training deep learning models.
NumPy	Numerical operations and array handling.
Pandas	Organizing experiment outputs and metric tables.
Scikit-learn	Dataset splitting and evaluation metrics.
Matplotlib	Plotting training curves and evaluation figures.
Google Colab	Notebook-based experiment environment.
PlantVillage	Source dataset for potato leaf images.

4.3 Deep Learning Models

4.3.1 Custom Convolutional Neural Network

A Custom CNN was developed from scratch.

Its architecture used convolutional and max-pooling layers to extract image features, followed by fully connected layers and a three-unit Softmax output layer.

4.3.2 MobileNetV2

MobileNetV2 was used as a transfer-learning model with ImageNet pretrained weights.

The original classification head was replaced with a task-specific classification head.

The base network was initially frozen while the new classification head was trained. Subsequently, the last 20 layers were unfrozen for fine-tuning with a lower learning rate.

4.4 Dataset and Input Configuration

The filtered dataset contained 2,152 images from three potato classes.

Images were converted to RGB format and resized to 224×224 pixels.

Pixel values were normalized to the range $[0, 1]$. The batch size was 32.

5 Internship Activities

5.1 Research Problem Selection

The technical work focused on classifying potato leaf images into three categories:

- Potato Early Blight
- Potato Late Blight
- Potato Healthy

This task was selected to explore how deep learning models can identify disease-related visual patterns from leaf images.

5.2 Dataset Collection and Preparation

The PlantVillage dataset was used as the source of image data.

Potato-related images were selected from the original dataset, resulting in 2,152 images distributed across the three target classes.

Table 3: Distribution of the Filtered Potato Leaf Dataset

Class	**Images**	**Percentage**
Potato Early Blight	1,000	46.47%
Potato Late Blight	1,000	46.47%
Potato Healthy	152	7.06%
Total	**2,152**	**100%**

The class distribution shows that the Healthy category was underrepresented compared with the two disease classes.

5.2.1 Sample Dataset Image

Figure 2 presents the sample dataset image used in the research.

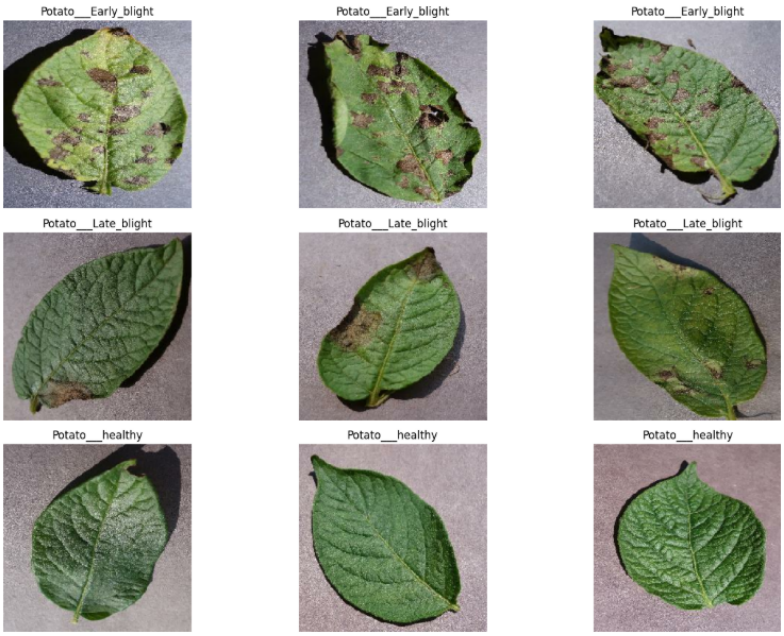


Figure 2: Sample potato leaf image from the dataset.

5.3 Image Preprocessing

The following preprocessing steps were applied:

1. Selected only the three potato leaf categories.
2. Converted images to RGB format.
3. Resized all images to 224×224 pixels.
4. Normalized pixel values using $x_{norm} = x/255$.
5. Split the dataset using a stratified approach with random seed 42.

Table 4: Dataset Split

Subset	**Images**	**Percentage**
Training	1,506	70%
Validation	323	15%
Testing	323	15%
Total	**2,152**	**100%**

Random horizontal flipping was used as the main augmentation technique for the training data.

No augmentation was applied to the validation or test data.

5.4 Model Development and Training

5.4.1 Custom CNN Implementation

The Custom CNN consisted of three convolutional blocks with 32, 64, and 128 filters, respectively.

Each block used a 3×3 convolution with ReLU activation and a 2×2 max-pooling layer.

The feature maps were flattened and passed through a 128-unit dense layer, a dropout layer with rate 0.5, and a three-unit Softmax output layer.

Table 5: Custom CNN Architecture

Layer	**Configuration**
Input	$224 \times 224 \times 3$
Conv2D + MaxPooling	32 filters, 3×3
Conv2D + MaxPooling	64 filters, 3×3
Conv2D + MaxPooling	128 filters, 3×3
Flatten	Feature vector
Dense	128 units, ReLU
Dropout	0.5
Output	3 units, Softmax

The Custom CNN used Adam with a learning rate of 0.001, sparse categorical cross-entropy loss, batch size 32, and a maximum of 15 epochs.

Early stopping used a patience value of 10, and the checkpoint monitored validation accuracy.

5.4.2 MobileNetV2 Implementation

MobileNetV2 was initialized with ImageNet pretrained weights, with its original classification head removed.

Global average pooling, a 128-unit ReLU dense layer, dropout of 0.5, and a three-unit Softmax output layer were added.

The first stage froze the MobileNetV2 base and trained the new classification head for up to 10 epochs using Adam with a learning rate of 0.001.

The second stage unfroze the last 20 layers and fine-tuned the model for up to five epochs using a learning rate of 0.00001.

5.5 Training and Validation Performance

The training and validation curves provide a visual representation of model learning during the training process.

They help illustrate changes in accuracy and loss across epochs and provide insight into the model's convergence.

5.5.1 Custom CNN Training Performance

Figure 3 shows the training and validation accuracy of the Custom CNN.

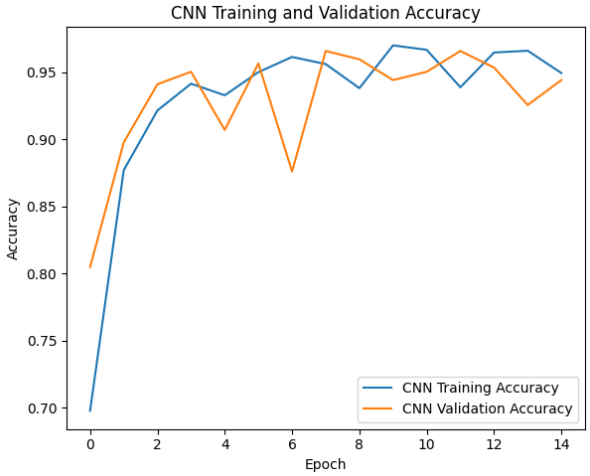


Figure 3: Custom CNN training and validation accuracy.

Figure 4 shows the training and validation loss of the Custom CNN.

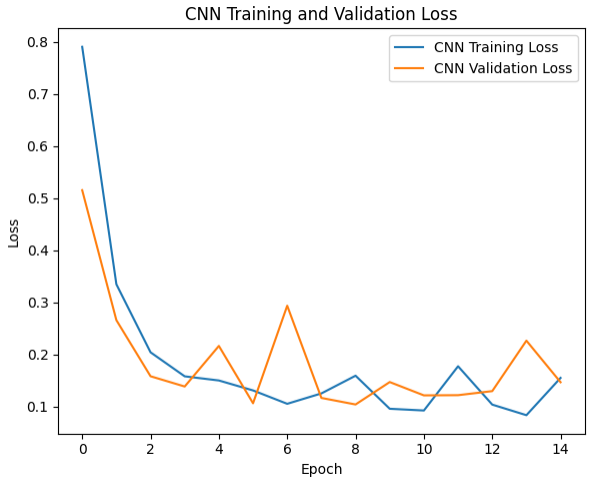


Figure 4: Custom CNN training and validation loss.

5.5.2 MobileNetV2 Training Performance

Figure 5 shows the training and validation accuracy of MobileNetV2.

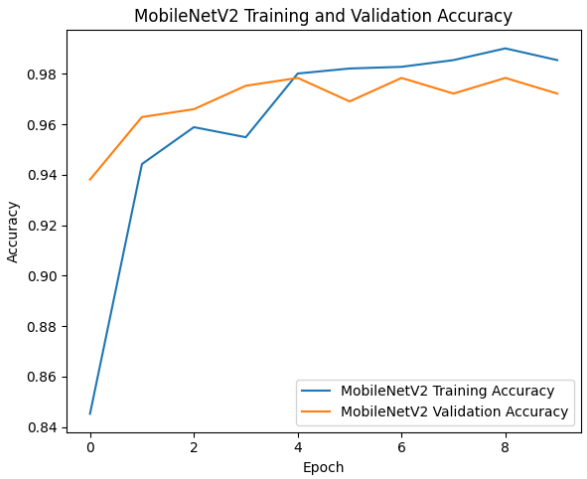


Figure 5: MobileNetV2 training and validation accuracy.

Figure 6 shows the training and validation loss of MobileNetV2.

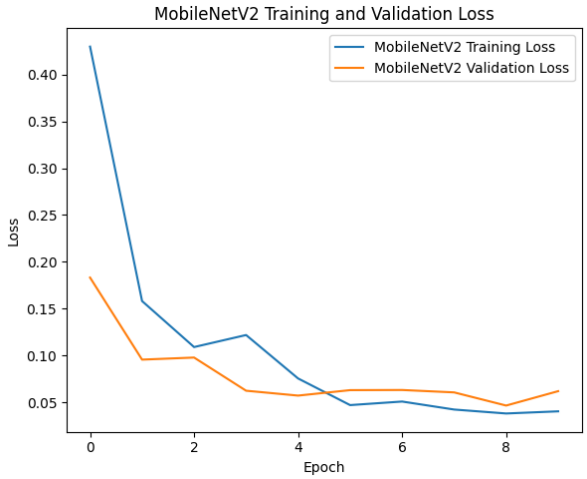


Figure 6: MobileNetV2 training and validation loss.

5.6 Model Evaluation

Both models were evaluated on the same held-out test set of 323 images.

The evaluation included accuracy, macro precision, macro recall, macro F1-score, class-wise metrics, confusion matrices, and one-vs-rest ROC-AUC scores.

5.6.1 Overall Performance Comparison

Table 6: Corrected Overall Model Comparison

Metric	Custom CNN	MobileNetV2
Accuracy	0.9412	0.9752
Macro Precision	0.9045	0.9687
Macro Recall	0.9332	0.9454
Macro F1-score	0.9178	0.9564

The results indicate that MobileNetV2 achieved higher overall accuracy, macro precision, macro recall, and macro F1-score than the Custom CNN.

5.6.2 Confusion Matrix Analysis

A confusion matrix provides a detailed comparison between the actual class labels and the predicted class labels.

The class order in both matrices is:

1. Early Blight
2. Late Blight
3. Healthy

Rows represent true labels, while columns represent predicted labels.

Custom CNN Figure 7 presents the confusion matrix of the Custom CNN.

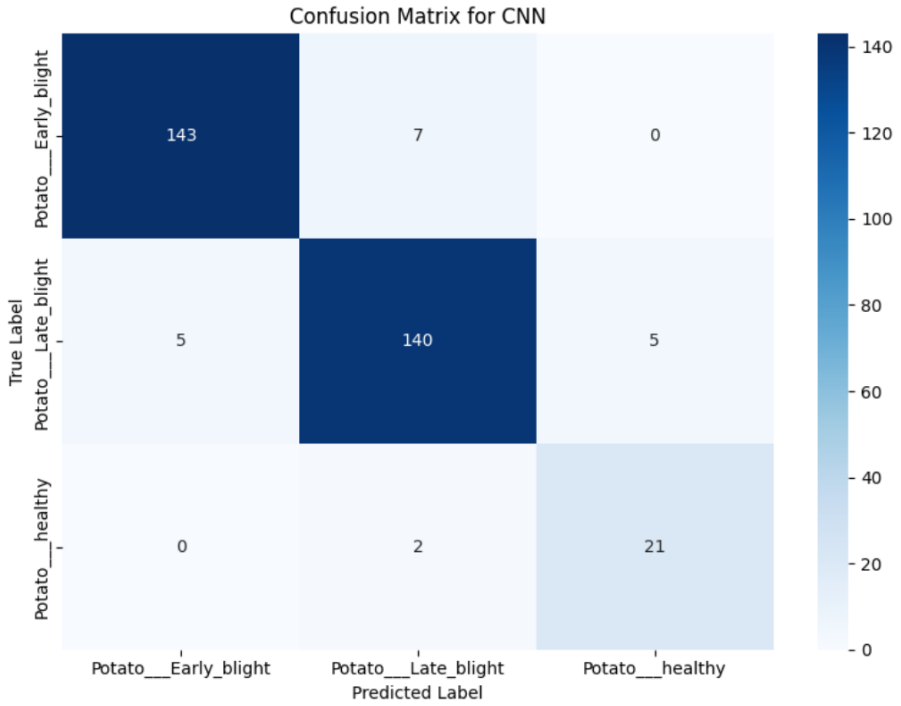


Figure 7: Confusion matrix of the Custom CNN.

The Custom CNN correctly classified 304 out of 323 test images, achieving an accuracy of 94.12%.

MobileNetV2 Figure 8 presents the confusion matrix of MobileNetV2.

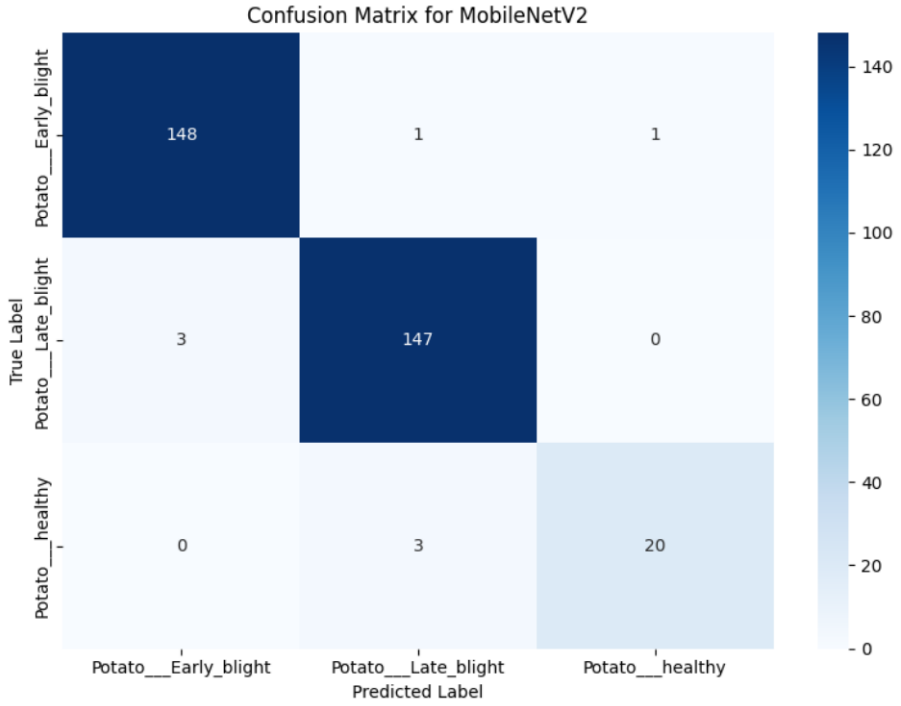


Figure 8: Confusion matrix of MobileNetV2.

MobileNetV2 correctly classified 315 out of 323 test images, achieving an accuracy of 97.52%.

5.6.3 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were used to examine the classification performance of each class under a one-vs-rest evaluation approach.

The Area Under the Curve (AUC) summarizes the ability of a model to distinguish a particular class from the remaining classes across different thresholds.

Custom CNN Figure 9 presents the one-vs-rest ROC curves for the Custom CNN.

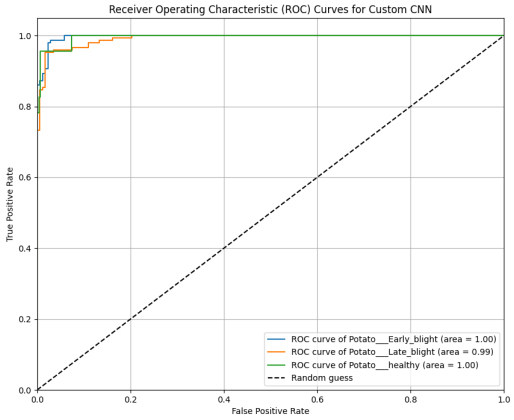


Figure 9: One-vs-rest ROC curves for the Custom CNN.

MobileNetV2 Figure 10 presents the one-vs-rest ROC curves for MobileNetV2.

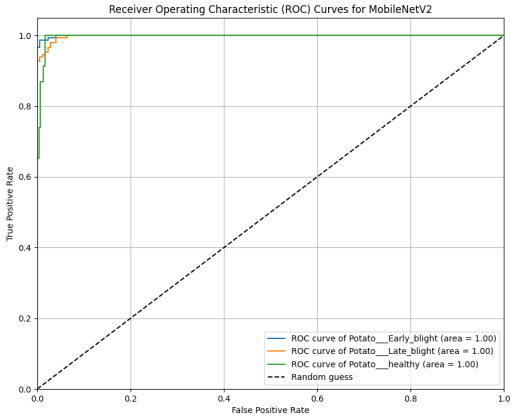


Figure 10: One-vs-rest ROC curves for MobileNetV2.

5.6.4 Class-wise Performance

Table 7: Custom CNN Classification Metrics

Class	Precision	Recall	F1-score	Support
Early Blight	0.9662	0.9533	0.9597	150
Late Blight	0.9396	0.9333	0.9365	150
Healthy	0.8077	0.9130	0.8571	23
Macro average	0.9045	0.9332	0.9178	323
Weighted average	0.9426	0.9412	0.9416	323

Table 8: MobileNetV2 Classification Metrics

Class	Precision	Recall	F1-score	Support
Early Blight	0.9801	0.9867	0.9834	150
Late Blight	0.9735	0.9800	0.9767	150
Healthy	0.9524	0.8696	0.9091	23
Macro average	0.9687	0.9454	0.9564	323
Weighted average	0.9751	0.9752	0.9750	323

5.6.5 AUC Comparison

Table 9: Recorded One-vs-Rest AUC Scores

Class	Custom CNN	MobileNetV2
Potato Early Blight	1.00	1.00
Potato Late Blight	0.99	1.00
Potato Healthy	1.00	1.00
Mean AUC	0.9967	1.00

The recorded AUC values should be interpreted alongside the confusion matrices and the limited, imbalanced test set.

6 Technical Learning

6.1 Image Preprocessing

I gained practical experience in preparing image datasets for deep learning.

This included class filtering, image resizing, RGB conversion, normalization, stratified splitting, and applying augmentation only to the training data.

6.2 CNN Architecture and Feature Extraction

The Custom CNN experiment helped me understand how convolutional layers learn spatial features from images.

Pooling layers reduce feature-map dimensions, while dense layers combine extracted features for final classification.

6.3 Transfer Learning and Fine-Tuning

The MobileNetV2 experiment demonstrated how a pretrained network can be adapted to a new classification task.

Freezing the base network allows the new classification head to learn task-specific outputs first. Fine-tuning selected layers then allows part of the pretrained feature extractor to adapt to the target dataset.

6.4 Model Evaluation

I developed a clearer understanding of why accuracy alone is not sufficient for an imbalanced multiclass dataset.

Precision, recall, and F1-score provide class-level information, while a confusion matrix shows which classes are being confused.

ROC-AUC provides another view of class discrimination across thresholds.

6.5 Experiment Tracking and Reproducibility

Using a fixed random seed, consistent input dimensions, and the same test set helped make the model comparison more controlled.

I also learned the importance of checking that confusion matrix counts, classification reports, and overall metrics agree before preparing a final report.

7 Problems Observed

7.1 Class Imbalance

The dataset contained 1,000 images in each disease class but only 152 Healthy images.

The test set consequently contained 150 Early Blight images, 150 Late Blight images, and 23 Healthy images.

This imbalance can influence model learning and makes the Healthy-class results particularly important to inspect.

7.2 Confusion Between Similar Classes

Some images from one disease category were predicted as another category.

Early Blight and Late Blight can contain visual patterns that are difficult for a model to distinguish consistently.

7.3 Limited Real-World Diversity

The experiment used PlantVillage images.

Such a dataset is useful for controlled experimentation, but its image conditions may not represent the range of lighting, backgrounds, camera quality, leaf positions, and disease severity found in field photographs.

7.4 Risk of Overfitting

The training metrics, particularly during MobileNetV2 fine-tuning, were very high.

A small difference between training and validation loss does not by itself establish generalization to new environments.

Independent field images would be needed to assess real-world performance.

7.5 Metric Consistency and Result Verification

During report preparation, the initially documented overall metrics did not match the confusion-matrix counts.

Recalculating the metrics from the final matrices resolved this inconsistency.

This highlighted the importance of using one verified source of predictions for all reported evaluation measures.

7.6 Limited Scope

The experiment considered only three potato leaf categories and did not include external field validation, explainability analysis, or deployment as a farmer-facing application.

8 Proposed Solutions

8.1 Addressing Class Imbalance

The Healthy class should be expanded by collecting more representative images.

Class weighting, balanced sampling, or carefully controlled oversampling could also be investigated.

Any balancing method should be applied only to the training data, not to the test set.

8.2 Improving Generalization

Future experiments should include images captured under different field conditions.

Data augmentation could be expanded with realistic variations in brightness, contrast, rotation, and scale, provided these transformations preserve disease characteristics.

8.3 Improving Evaluation Reliability

A single saved set of test predictions should be used to generate the confusion matrix, classification report, and ROC-AUC results.

Automated checks should verify that the row totals equal the class supports and that the diagonal sum equals the number of correct predictions.

8.4 Further Model Investigation

Additional architectures such as EfficientNet, ResNet, and DenseNet could be evaluated under the same data split and evaluation procedure.

Hyperparameter tuning could also be performed systematically.

8.5 Explainability and Deployment

Grad-CAM or a similar explainability method could be used to visualize image regions that influence model predictions.

After evaluation on independent field images, a lightweight web or mobile interface could be considered as a future application.

The current experiment should not be treated as a validated field-diagnosis tool.

9 Reflection on Professional Skills

9.1 Research and Analytical Skills

The internship strengthened my ability to break a broad technical problem into smaller stages: understanding the task, preparing data, selecting models, conducting experiments, evaluating results, and interpreting limitations.

I learned to avoid relying on a single performance number and to examine class-wise behavior.

9.2 Problem-Solving Skills

The inconsistency between earlier metric summaries and the confusion matrices required careful checking.

Recomputing the metrics from the final prediction counts helped me understand the importance of verification and evidence-based reporting.

9.3 Technical Communication

Preparing this report improved my ability to describe a technical workflow in a structured way.

Tables, equations, plots, and confusion matrices helped communicate the experiment more clearly than unstructured notes.

9.4 Time Management and Independent Work

The project required coordinating data preparation, model training, evaluation, and documentation.

Working through these stages helped me practice planning tasks, monitoring experiment progress, and keeping records of configuration choices and results.

9.5 Professional Conduct and Future Development

The internship provided experience in working on an assigned research task under supervision.

It also showed me areas where I need further development, including testing models on external datasets, improving experiment tracking, learning more about explainable AI, and communicating limitations clearly.

10 Conclusion

This report has presented my individual internship experience at Authentic Four Technology, where my primary technical activity focused on machine learning research for potato leaf disease classification.

The work used a filtered PlantVillage dataset containing 2,152 images across Early Blight, Late Blight, and Healthy categories.

A Custom CNN trained from scratch and a MobileNetV2 transfer-learning model were implemented and evaluated on the same 323-image test set.

The corrected Custom CNN confusion matrix was:

$$\begin{bmatrix} 143 & 7 & 0 \\ 5 & 140 & 5 \\ 0 & 2 & 21 \end{bmatrix}$$

This gives 304 correct predictions and an accuracy of 94.12%.

Its macro precision, macro recall, and macro F1-score were 90.45%, 93.32%, and 91.78%, respectively.

The corrected MobileNetV2 confusion matrix was:

$$\begin{bmatrix} 148 & 1 & 1 \\ 3 & 147 & 0 \\ 0 & 3 & 20 \end{bmatrix}$$

This gives 315 correct predictions and an accuracy of 97.52%.

Its macro precision, macro recall, and macro F1-score were 96.87%, 94.54%, and 95.64%, respectively.

MobileNetV2 therefore achieved stronger overall test accuracy, macro precision, macro recall, and macro F1-score in this corrected comparison.

The recorded class-wise AUC values were 1.00, 0.99, and 1.00 for the Custom CNN, and 1.00 for all three MobileNetV2 classes.

These AUC values should be interpreted alongside the confusion matrices and the limited, imbalanced test set.

Overall, the internship helped me apply deep learning concepts to an image-classification research problem and improve my skills in data preparation, model development, evaluation, result verification, and technical reporting.

Further work with balanced data and independent field images is required before considering practical agricultural use.

11 References

1. S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in Plant Science*, vol. 7, 2016. Available: https://doi.org/10.3389/fpls.2016.01419.
2. S. P. Mohanty, PlantVillage Dataset. Available: https://github.com/spMohanty/PlantVillage-Dataset.
3. M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
4. TensorFlow, "TensorFlow Documentation." Available: https://www.tensorflow.org/.
5. Keras, "Keras Documentation." Available: https://keras.io/.
6. Scikit-learn, "Scikit-learn Documentation." Available: https://scikit-learn.org/stable/.
7. Authentic Four Technology, official website. Available: https://www.authenticfour.com/. Accessed: [Access Date].